can also be dispersed in water to form cubosomes. Cubosomes can incorporate enzymes and stabilize their native state[11]. The particle size can be varied from about a hundred to up to several thousand nanometers. The incorporation of glucose oxidase to reduce oxygen content or of lactase to digest lactose might be considered as potential applications of cubosomes.

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Review

Unraveling the secret of meat flavor

Gilbert Idolo Imafidon and Arthur M. Spanier

Meat flavor arises from the complex interactions among amino acids, peptides, sugars, thiamine, metabolites of nucleotides, lipids and products of lipid oxidation. Previous research on meat flavor has shown numerous volatile compounds to be involved in meat aroma but the identities of specific mixtures of compounds contributing to individual meat flavors are not fully known. Factors that exert significant influence on meat flavors such as the nutritional status of the animal, the animal species and the temperature at which the meat is cooked are discussed. Lastly, it is suggested that the analytical procedures and the interpretation of the data arising from such analyses of meat flavor volatiles may be hindering rather than contributing to our understanding of meat flavor.

The processes by which the relatively bland, metallic, slightly salty taste[1] and serum-like odor of raw meat becomes distinctly flavored have been investigated for many decades. Despite all this effort, the processes of flavor development and deterioration in meat or muscle foods are not well understood. It is known that perceived meat flavor arises from a complex reaction of flavor precursors that interact or degrade during cooking and/or storage to produce compounds that are responsible for characteristic meat flavors. Not only does each muscle food, including that from beef cattle, pigs, goats and sheep, possess its own distinct flavor, but these flavors can be intensified or altered by different methods of cooking and also by cooking to different end-point temperatures. Although over 1000 volatile compounds in meat have been identified to date, the exact contribution of each compound to meat flavor is not yet known. Also, the vast number of possible interactions of individual flavors to produce the perceived taste of meat are unknown. It is because of all these unknown and poorly understood events that more research designed to examine and eventually reveal the secrets of meat flavor should be carried out.

Flavor perception in humans involves three distinct sensory systems (olfactory, gustatory and trigeminal) responding to food components[2]. The basic tastes of sweet, salty, sour, bitter and umami stimulate gustatory receptors within the oral cavity in response to the presence of soluble compounds[3]. Umami describes the savoriness, deliciousness or succulence of a food and is considered by the Japanese to represent a fifth taste. Volatile compounds stimulate the olfactory sensory organs in the roof of the nasal cavity. On the other hand, the trigeminal sense, a system that is dependent on free nerve endings in the eyes, nose and mouth, is stimulated by the presence of coolness, heat, astringency, acridness and pungency[3].

Meat flavor development and deterioration has been shown to be affected by several antemortem and postmortem factors[4,5] (Box 1). Although several antemortem

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Box 1. Some factors that affect meat flavors[3]

Antemortem

Age

Species/breed

Sex

Nutritional status

Stress level

Fat level

Fat composition

Postmortem

Manner of slaughter

Carcass handling

Aging increases hydrolytic activity and levels of products of the activity of endogenous and/or microbial proteinases, lipases and glycosidases

Manner of cooking:

- Wet versus dry
- Convection versus microwave
- Rate of heating
- Final end-point cooking temperature
- Final fat levels and composition

Manner of storage after cooking:

- Lipid → autoxidation → off-flavor
- Peptides → decreased hydrophilicity → increased off-flavor

factors such as nutritional status, species and age of the animal are of significance to perceived meat flavor, the most important factors are typically those that develop during postmortem aging and also during cooking and processing periods[5-8]. This review briefly discusses the complexity of meat flavor from the point of view of flavor precursors and flavor compounds in muscle foods. The effects of species, dietary regime of the animal and cooking temperature as well as method of cooking and the presence of non-meat ingredients on the development and deterioration of meat flavor components are also discussed.

Meat sources

Meat or muscle foods are typically obtained from a wide variety of animals such as beef cattle, pigs, sheep, goats, poultry, wild game and fish. In general, beef, pork and poultry meat are commonly consumed in the Western diet. On the other hand, goat meat is consumed by more individuals than any other single type of red meat[9], but the per capita consumption is small. The relative consumption of the various types of meat, however, depends on geographic location, socioeconomic status, religious dietary laws and the ability of the animal breed to survive and reproduce under the local geographic and climatic conditions.

Meat structure

Muscle cells are multinucleated, mitochondria-rich structures that are highly compact and complex in their structural organization. Muscle contains contractile elements that constitute the bulk of its protein. The contractile elements are surrounded by the highly evolved sarcoplasmic/lysosomal membrane system[2]. The close location of the contractile proteins to the sarcoplasmic/lysosomal membrane system places them in the optimal location for hydrolysis by lysosomal hydrolases. The structure of meat also allows only the surface of the muscle, especially the phospholipids of muscle membranes, to be exposed directly to molecular oxygen, a major catalyst in the generation of compounds that contribute to the off-flavors typically considered to be undesirable by the consumer[3]. These anatomical features have significant consequences on meat flavor development and deterioration[7]. It is now generally recognized that any process that entails the cooking, grinding or restructuring of muscle enhances not only the development of desirable flavor but also the development of warmed-over flavor or meat flavor deterioration.

Muscle to meat

The development of meat flavor precursors during the postmortem conditioning period arises from the cascade of biochemical events summarized in Fig. 1, and is similar in all meats. Ischemia is defined as no or low blood flow, similar to the events associated with slaughter. Following slaughter, intracellular ATP (adenosine triphosphate) concentration decreases causing the lysosomal membrane to rupture, releasing proteases, lipases and glycosidases, which further weaken membrane integrity and cellular function[8]. This process is referred to in Fig. 1 as cell death, and is defined as the cessation of normal metabolic function and activity in the cell or tissue. Acidosis increases during the postmortem period, due to the anaerobic conditions associated with the diminished blood circulation, which lowers the intramuscular pH to a level that is optimal for the activity of several lysosomal thiol proteinases[10]. As muscle is primarily proteinaceous, and proteolysis increases during postmortem aging, it is reasonable to conclude that muscle represents a remarkable pool from which flavor precursors of protein origin may be generated[7,10,11].

Meat flavor precursors

Raw meat has a mild, serum-like odor. It is only after cooking that meat develops its associated flavor. The flavor precursors of meat include non-volatile components (peptides, amino acids, some organic acids, sugars, metabolites of nucleotides, water, thiamine and lipids), and products of lipid oxidation/degradation, which constitute a major portion of the volatile components[5,12-16]. The Maillard reaction products (MRP), formed by the reaction of reducing sugars and amino acids or peptides, constitute another major category of volatile components. The biochemistry of these flavor precursors and methods for their analysis have been discussed previously[5,13,16].

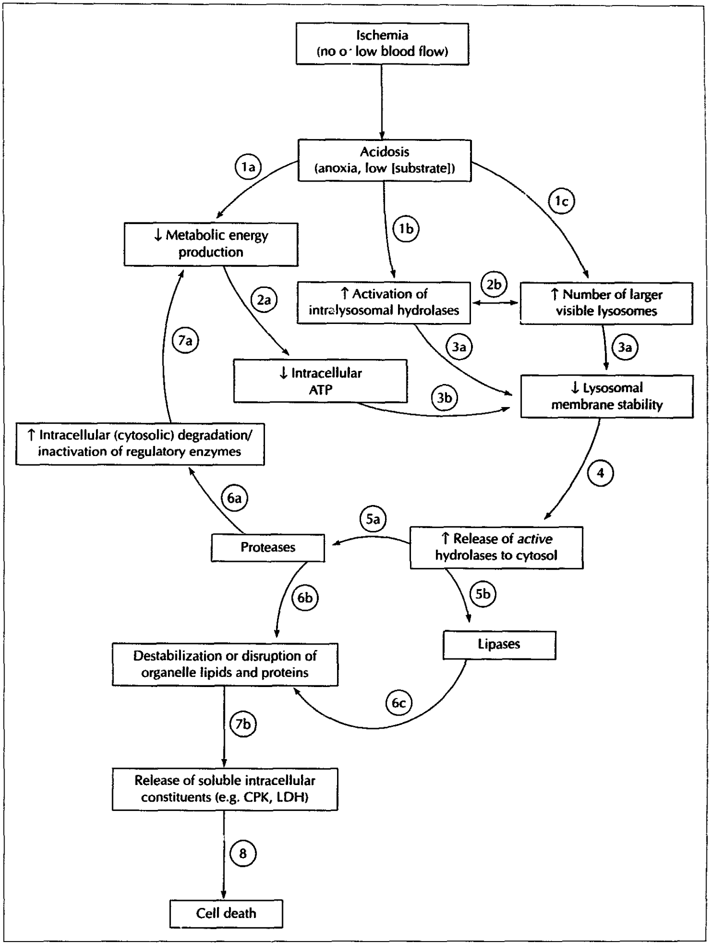

Fig. 1

Early biochemical events associated with the postmortem conditioning period of muscle. Numbers refer to the speculated sequence of events. When numbers are followed by letters (e.g. 1a, 1b, 1c, etc.) the events are thought to occur simultaneously. LDH and CPK denote lactate dehydrogenase and creatine phosphokinase, respectively. Arrows indicate increases (↑) or decreases (↓). Square brackets [ ] indicate concentration.

Several differences in addition to variations in the pH of raw beef, pork and lamb have been reported. These include differences in the content of acidic, basic and sulphur-containing amino acids and in the low levels of dipeptides, such as anserine and carnosine, in lamb as compared with beef and pork[14]. This suggests that the volatiles of processed lamb, pork and beef will differ qualitatively and/or quantitatively[14]. Certain amino acids contribute to the sweet taste of meat[16], whereas peptides and hypoxanthine contribute to bitter and umami tastes. Some amino acids and other acids, such as lactic, inosinic, orthophosphoric and succinic acids, also contribute acidic flavor to meat[16]. Sugars and sodium salts of glutamic and aspartic acids furnish sweet and salty characters to meat flavor, respectively[17,18]. Metabolites of nucleotides are potential sources of reducing sugars, such as free ribose and ribose phosphate[15]. These sugars and the reducing sugars from glycolysis are important in the generation of MRP. The proposed mechanisms of formation of flavor precursors have been discussed by various workers[2,6,8,19,20].

There is still some debate over the flavor that distinguishes the meat of different species[21]. The essential precursors of meaty flavor are attributable to the non-volatile, water-soluble, low molecular weight components of meat[13]. Species-specific flavor is attributable mainly to the volatile components[5–7,13,14]. It is interesting to note that even within the same species, the lipid-derived flavors are affected by dietary regime, as will be discussed below.

Although there is considerable literature on meat flavor precursors, little is known about the origin of the non-volatile flavor precursors. For example, the parent protein of BMP (beefy meaty peptide), an octapeptide found naturally in raw beef, which is known to contribute to the savoriness of the cooked meat[2], is unknown. In addition, there may be a similar peptide or peptides with the ability to potentiate flavor(s) in other muscle foods including poultry meat, lamb and pork. The determination of the presence of these flavor peptides and their parent protein(s) *in situ* is a matter of considerable interest and represents a challenging area for future research.

Non-volatile flavor compounds in meat

During the postmortem period, the levels of flavor precursors in meat alter significantly[10,22]. Many of these changes are due to enhanced hydrolytic activity. These postmortem events produce a rather large pool of reactive flavor components, which interact during cooking to form additional flavor notes. Regardless of the history or origin of the meat, because it is stored and reheated, both desirable and undesirable flavors develop from glycolysis, proteolysis, lipolysis, oxidation, pyrolysis and the interaction of the resultant products[5,12,14]. Lipid autoxidation is a prime cause of the production of off-flavor components in meat[7,19]. Ribonucleotides, such as inosine 5′-monophosphate and guanosine 5′-monophosphate, play important roles in the flavor of meat owing to their ability to potentiate the meaty flavor and suppress the sulfury, fatty, burnt, starchy, bitter and hydrolyzed vegetable type flavors in meat[22]. Inosinic acid, in addition to its potentiating action, has been reported to be essential for the development of meaty aroma during cooking[22,23].

Increased amounts and types of proteinaceous components have been shown to develop with increasing postmortem aging[10]. Current evidence suggests that in acid extracts of beef, two subclasses of peptides (1800 Da) are found both during postmortem aging and at different end-point cooking temperatures. One of these peptide subclasses is rich in hydrophilic residues and the other rich in hydrophobic residues. Hydrophilic residues are associated with desirable flavor, whereas hydrophobic residues are associated with undesirable or off-flavors[10]. These peptides are thought to be produced in the raw meat by proteolytic fragmentation[10,20]. On the other hand, if precooked meat is stored (4°C), there is a decline in the presence of the hydrophilic residues originally present, but no major change in the hydrophobic residues[2,6,11]. The loss of hydrophilic residues with storage is thought to be a function of the fragmentation of parent hydrophilic residues, probably due to reactions with free radicals produced by lipid oxidation. The loss of hydrophilic peptides is closely related to the changes in flavor of the stored meat sample; upon storage there is generally an increase in undesirable bitter and sour notes and a decline in the more desirable cooked beef/brothy note[24].

Similar changes in flavor-producing compounds have been reported in goat[25] and sheep meat[22], pork and chicken during storage[23,26]. In sheep and goat meat, inosinic acid was the most predominant nucleotide at 6–8 h postmortem[22]. After cooking at 60, 80, 100 or 120°C, inosinic acid levels decreased, unlike those of other nucleotides. At 120°C, the thermal degradation of inosinic acid proceeded at a significant rate, but at temperatures below 100°C major changes occurred because of the activity of phosphomonoesterase during the initial stages of heating[22]. In earlier studies, inosinic acid, inosine and guanylic acid were also described as important components contributing to the complexity of chicken flavor[22].

Although chemical changes to protein during cooking have been well documented, few studies deal with the role of peptides in cooked-meat flavor. The potential of peptides such as BMP and similar flavor peptides from red meat has been discussed and is still a subject of current investigation[2,10,11]. The optimal activity of some proteolytic enzymes in beef occurs at ~68°C – the transition temperature at which many of the major changes in fresh meat proteins occur[10]. Increasing amounts and types of peptides and amino acids in beef were shown to occur with increasing end-point cooking temperature in the range 51.7–76.7°C (125–170°F)[10].

Volatile flavor compounds in meat

Several analytical techniques are available for monitoring the presence of volatile meat flavor compounds[26,27-30]. These methods have been used to isolate and characterize many odorous compounds such as hydrocarbons, aldehydes, ketones, alcohols, carboxylic acids, esters, lactones, furans, pyridines, pyrazines, alkylphenols, thiols, thiophenols, thiazoles, other nitrogen compounds, halogenated compounds, and sulfur-containing and other miscellaneous compounds that contribute to flavor[12,27-29]. The concentrations and threshold values of these flavor-producing components determine how they are perceived and, thereby, their desirability in meat flavor.

The reaction of sugars with amino acids is known as the Maillard reaction, and the compounds produced are known as Maillard reaction products (MRP). MRP are considered to have a vital role in the production of flavor in meat[1,14]. A significant proportion of the MRP that produce meaty-like flavors contain sulfur. Meaty-like flavors are also obtained from other heterocyclic compounds formed during Maillard reactions. Meat flavors of prepared foods rely heavily on MRP, although little is known about their formation and influence on the overall sensory perception and flavor of meat[5,6]. Thus, the mechanism of MRP formation, the specific content of MRP in meat and the influence of MRP on sensory perception should represent a major area of investigation in meat and food science[6].

Other effectors of meat flavor

Descriptors

Flavor descriptors for meat from several species of meat animal have been reviewed recently[31]. Table 1 summarizes the majority of the descriptors in the meat flavor lexicon. There are many similarities in the flavor descriptors used for the various meat species. For example, the four basic tastes of sweet, sour, salty and bitter can be found at various levels or intensities in meats from most species[5,31]. Like the seminal report on the descriptors of flavor deterioration in beef[33], standardized descriptive terms for species-specific flavor[31,32] should be designed and adopted. In particular, the literature contains no information on flavor descriptors for goat meat other than the general term of 'goaty flavor' on some measurable scale (e.g. least favorable to most favorable conditions)[34]. Although there are differences between the flavor of goat meat and that of beef and/or sheep meat, the overall acceptability is highly individual.

Table 1. Flavor descriptors for various species of raw and cooked meat[a]

Meat descriptor	Associated product/process
Beef	
Beefy	Matured cooked beef muscle products
Brothy/meaty	Drippings from roasted meat; characteristic of all meats
Serum/raw	Raw lean beef
Browned caramel	Outside of grilled or broiled beef, seared but not blackened/burnt
Grainy/cowy	Cow meat and/or beef in which grain/feed character is detectable
Cooked liver	Cooked organ meats such as liver
Cardboardy	Slightly stale beef (refrigerated for a few days only); associated with wet cardboard and stale oils and fats
Painty	Rancid oil and fat
Goat	
Goaty	Cooked goat meat
Lamb	
Gamey/muttony	Muscle meat from wild game or older lambs
Fatty	Cooked lamb fat
Bloody/serum	Raw lean meat
Browned caramel	Outside of grilled or broiled lamb, seared but not burnt
Cooked liver	Cooked organ meats such as liver
Meaty	Cooked muscle meat
Musty/herby	Wet soil or mulch and dried herbs, such as rosemary or thyme
Pork	
Porky	Cooked pork meat
Sex taint/boar odor	Cooked pork meat
Fatty	Cooked pork fat
Bloody	Probably similar in all red meat (serum/raw)
Poultry	
Chickeny/poultry	The chickeny/poultry flavor is unique to poultry and is associated with cooked white chicken muscle
Meaty	Cooked dark chicken muscle
Browned	Outside of grilled or broiled chicken, seared but not burnt
Brothy	Chicken stock
Livery/organy	Cooked organ meat such as liver

[a]Taken from Refs 31 and 32

Presence of other compounds

Although there has been much discussion about the general cooked-meat flavor descriptors, the origins of the precursors and the exact chemistry of the compounds causing the characteristic species-related flavors of meats remain complex issues. For instance, the distribution of carbonyls varies with the lipid composition of the original meat, including beef, pork, lamb and poultry meat[16,29]. 16-Octadecenal, benzaldehyde, 2,3-octanedione and (*E,Z*)-2,4-decadienal may be responsible for the species-specific flavor notes in pork, whereas 2-hexanone and 3,3-dimethylhexanal have been identified in beef[28]. Also, unique pork-like or 'piggy' flavors of pork were considered to be caused by certain phenols in combination with a high concentration of 5-methylbutanoic acid[17]. Other examples of compounds identified in red meat are 4-methyloctanoic and 4-ethyloctanoic acids, which were found in abundant quantities to be characteristic of goat meat[15,17]. In addition to 4-methyloctanoic and 4-ethyloctanoic acids, sheep fats contain both the most and the highest concentrations of phenols and alkylphenols[17]. These acids and phenols are species-specific flavor compounds that are produced in cooked sheep meat, and are responsible for sheep-like or 'muttony' odor. In the same study, the phenols present in veal and pork were found to be similar except that pork contained 8 ppb of 3- and/or 4-ethylphenol. Studies on other species-specific compounds that contribute to red meat flavor abound in the literature[14,15,28], but the mechanism of the reactions of the mixtures of flavor-producing compounds that give the species-specific flavor note or bouquet *in vivo* have yet to be elucidated.

Feed composition

Certain differences in the levels of flavor-producing compounds in different muscle foods due to the diet fed to pigs, poultry, beef cattle, sheep and other animals have been reviewed[4,35]. In general, feeding high-energy grain diets to beef cattle, pigs and sheep produces a more acceptable or more intense flavor in red meats than low-energy forage or grass diets[36]. Feeding pigs with diets containing unsaturated fats significantly alters the fatty acid composition of subcutaneous fat without changing pork flavor or intensity[37]. Changes in fatty acid composition, especially in $C_{18:2}$ content, of the lean tissue seem to have little effect on the flavor of freshly cooked, lean meat. However, lean meat had higher concentrations of aldehydes, especially pentanal and hexanal, suggesting a high tendency for increased oxidation[36] and consequently the formation of off-flavor compounds in this meat after prolonged storage. In order to increase the degree of unsaturation of their muscle fat to be similar to that of pork fat, ruminants must be fed protected[38], unsaturated fats[35]. Such increased unsaturation results in a greater flavor change in lamb and beef than in pork. The concentrations of 41 out of 59 volatile compounds identified in beef were significantly influenced by diet[39]. Among these compounds, lactones and diterpenoids were correlated with the 'gamey/stale' off-flavor and 'roasted beef' flavor. Also, diet-related off-flavors in poultry meat occur very readily[39]. For example, feeding turkeys with diets containing highly unsaturated fats resulted in an off-flavor described as 'fishy'. Although the compound(s) responsible for the fishy off-flavor are not known, the undesirable fishy flavor in turkey could be eliminated by supplementing the feed with α-tocopherol[37]. Feeding chickens with omega-3 fatty acids [eicosapentaenoic acid (20 : 5ω-3) and docosahexaenoic acid (22 : 6ω-3)] derived from fish imparted a fishy flavor to chicken meat[40].

Cooking methods

Cooking methods and final internal temperatures have an important effect on the formation and stability of both volatile[10,20] and non-volatile compounds[10,15], and consequently on meat flavor[2,18,20,28,38,41]. For example, the formation of MRP is enhanced at higher cooking temperatures[42]. Several MRP have antioxidative properties[42,43] and, therefore, inhibit the rate of lipid autoxidation and deterioration of meat flavor[44]. Unfortunately, although elevated temperatures result in the formation of MRP with antioxidative properties, they also enhance the rate of lipid autoxidation. Enhanced myoglobin degradation[10] and the subsequent release of free iron[45] further augment the rate of lipid autoxidation. Major chemical changes, which include changes in enzyme activity and protein and amino acid composition, begin to occur at temperatures in the range 60–77°C (Ref. 10). Lipid autoxidation also increases within this temperature range. However, above this temperature range, the rate of lipid autoxidation decreases. This is attributed to the formation of MRP with antioxidative properties at the elevated temperatures[10]. Thermal processing of beef[10,19,20,46], pork[40], goat[22,25] and sheep[23,32] meat results, therefore, in changes in flavor components. Increased end-point temperatures in the range 65.5–82.2°C are associated with increased pork flavor and also increased concentrations of lipids, proteins and certain fatty acids[41]. Pork flavor is attributed to interaction between endogenous carbonyl–amine compounds to form MRP at the higher temperature.

Analytical techniques

There is an urgent need to re-evaluate the analytical techniques for examining meat flavor precursors and compounds[28]. Although many data exist on volatile compounds in meat, species-specific compounds have not been unambiguously established to aid quality assurance in meat products. Unfortunately, analytical techniques may alter the volatile flavor profile of the meat such that the volatiles present are not actually representative of the samples[28]. It has recently been demonstrated that the volatile flavor from cooked and cooked-stored ground beef are directly affected by, and related to, the purge temperature used to isolate the volatile compounds[28]. Data suggest that a true picture of a food's volatile composition and, therefore, its potential flavor, can only be appreciated through a thorough examination and understanding of the effect of temperature on the development and quantity of such volatiles.

Conclusions

Overall, the flavor of cooked meat can be defined as the interaction of a mixture of non-volatile compounds with taste properties, potentiators that enhance the inherent flavors of other compounds and the volatiles that give rise to the odor properties of meat[13,31].

Previous research designed to unravel the flavor of meat has thus far been unable to completely uncover the secret(s) of meat flavor. This is mainly because there are several classes of flavorful compounds, which are found in sufficiently high concentrations. These compounds need to be identified and assessed by sensory evaluation against relatively high levels and different types of background proteins. The important flavor compounds are more difficult to pinpoint in the case of beef, pork and poultry meat where the literature abounds with data about flavorful compounds. Numerous compounds can be identified at very low thresholds, yet at significant enough levels that it is difficult to select those that are important for flavor. It may be true that the flavor profiles of different species of meat consist of many different interacting compounds, whose individual attributes and contributions to desirable flavor have not yet been unraveled empirically. The emphasis on meat flavor research has traditionally been directed towards the volatile compounds, but there is a growing awareness that the non-volatile, water-soluble products of proteolysis may be important in determining the desirability of meat taste. It is only by continuing the current multidisciplinary research approaches[8,19], designed to correlate the fundamental chemical and physiological changes in meat with those changes in meat flavor[24,45], that our understanding of meat flavor development and meat flavor loss may be improved and the secret(s) of meat flavor revealed. With this knowledge we can design and manufacture meat and meat products that will satisfy the dynamic tastes of consumers.

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Conference Report: 4th Wartburg Aroma Symposium, Eisenach, Germany, by Alistair Paterson